Catálogo Completo de Distribuciones de Probabilidad

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Contents

Distribuciones Discretas

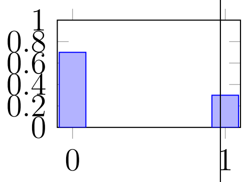
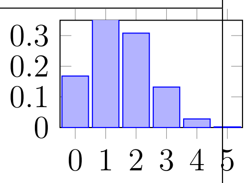
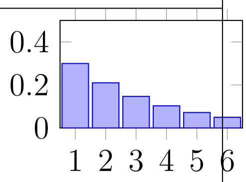
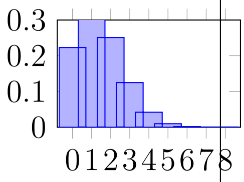
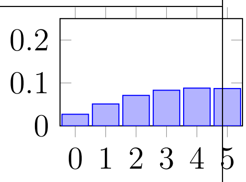
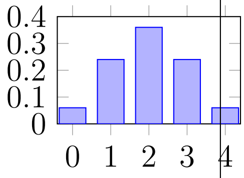
Distribución	**PMF**	**Media / Var / σ**	**Dominio**	**Gráfica**
Bernoulli	$P(X=1)=p, P(X=0)=1-p$	$E[X]=p,\ Var(X)=p(1-p),\ \sigma=\sqrt{p(1-p)}$	$X \in \{0,1\}$	
Binomial	$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$	$E[X]=np,\ Var(X)=np(1-p),\ \sigma=\sqrt{np(1-p)}$	$X \in \{0,\dots,n\}$	
Geométrica	$P(X=k)=(1-p)^{k-1}p$	$E[X]=\frac{1}{p},\ Var(X)=\frac{1-p}{p^2},\ \sigma=\sqrt{\frac{1-p}{p^2}}$	$X \in \{1,2,\dots\}$	
Poisson	$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}$	$E[X]=\lambda,\ Var(X)=\lambda,\ \sigma=\sqrt{\lambda}$	$X \in \{0,1,2,\dots\}$	
Binomial Negativa	$P(X=k)=\binom{k+r-1}{k}(1-p)^kp^r$	$E[X]=\frac{r(1-p)}{p},\ Var(X)=\frac{r(1-p)}{p^2}, \sigma=\sqrt{\frac{r(1-p)}{p^2}}$	$X \in \{0,1,2,\dots\}$	
Hipergeométrica	$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	$E[X]=n\frac{K}{N},\ Var(X)=n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$	$X \in \{\max(0,n-(N-K)),\dots,\min(n,K)\}$	

Table 1: Distribuciones discretas con PMF, medidas, dominio y gráficas.

Distribuciones Continuas

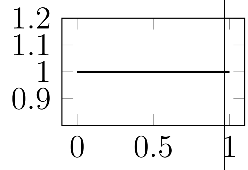
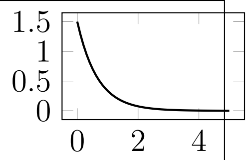
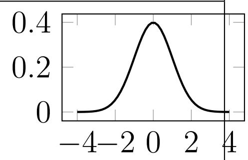
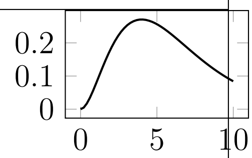
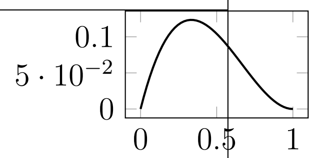
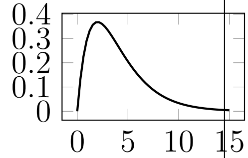
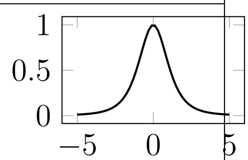
Distribución	**PDF**	**Media / Var / σ**	**Dominio**	**Gráfica**
Uniforme Continua	$f(x) = \frac{1}{b-a}$	$E[X] = \frac{a+b}{2}$, $Var(X) = \frac{(b-a)^2}{12}$, $\sigma = \frac{b-a}{\sqrt{12}}$	$x \in [a, b]$	
Exponencial	$f(x) = \lambda e^{-\lambda x}$	$E[X] = \frac{1}{\lambda}$, $Var(X) = \frac{1}{\lambda^2}$, $\sigma = \frac{1}{\lambda}$	$x \in [0, \infty)$	
Normal	$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$	$E[X] = \mu$, $Var(X) = \sigma^2$, σ	$x \in (-\infty, \infty)$	
Gamma	$f(x) = \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$	$E[X] = \alpha\beta$, $Var(X) = \alpha\beta^2$, $\sigma = \beta\sqrt{\alpha}$	$x \in (0, \infty)$	
Beta	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$E[X] = \frac{\alpha}{\alpha+\beta}$, $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$x \in (0, 1)$	
Chi-cuadrada	$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$	$E[X] = k$, $Var(X) = 2k$, $\sigma = \sqrt{2k}$	$x \in (0, \infty)$	
t-Student	$f(x) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	$E[X] = 0$, $Var(X) = \frac{\nu}{\nu-2}$, $\sigma = \sqrt{\frac{\nu}{\nu-2}}$	$x \in (-\infty, \infty)$	

Table 2: Distribuciones continuas con PDF, medidas, dominio y gráficas.